Fiscal Year:	FY 2011	**Task Last Updated:**	FY 01/31/2012
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Project Title:	Streptococcus pneumoniae Gene Expression and Virulence Potential in the Space Environment		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77555-1019	**Congressional District:**	14
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-02
Start Date:	10/01/2009	**End Date:**	08/31/2011
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-129 ; ISS NOTE: End date is now 8/31/2011 (previously 3/31/2011), per A. Chu/ARC (Ed., 3/14/2011) NOTE: End date is now 3/31/2011 (previously 9/30/2010) per A. Chu/ARC (10/20/10)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
Grant/Contract No.:	NNX09AW36G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Evidence has accumulated that extended exposure to the space environment has profound effects on human physiology. Experiments have shown that spaceflight reduces a wide spectrum of immune functions. Immunodeficiency can increase susceptibility to cancer and infection. Multiple studies have shown that the space environment and ground-based analog models of microgravity enhance the virulence potential of both frank and opportunistic microbial pathogens. Thus, with a combination of decreasing host immune function and increased bacterial virulence, normally commensal bacteria can emerge as opportunistic pathogens capable of causing severe disease. We have been investigating the virulence of S. pneumoniae (also known as the pneumococcus), a commensal organism and that has been cultured from the nasopharynx of crewmembers prior to flight. Our recent studies have shown that the virulence properties of this opportunistic pathogen increase under spaceflight and simulated microgravity conditions. Experiments were conducted on recent shuttle flights - STS 118 and STS 123 in August, 2007 and March, 2008, respectively. These experiments have shown that exposure to the space environment results in a major re-programming of gene and protein expression and enhancement of adherence of S. pneumoniae to epithelial cells. Further, S. pneumoniae grown in land-based microgravity analog models and on board STS 123 have been significantly more virulent in a mouse intraperitoneal infection model than are paired controls. Collectively, these results provide strong evidence that microgravity environments increase the virulence of this opportunistic pathogen. Understanding virulence enhancement after bacterial growth in the space environment is critical to the long-term goal to devise countermeasures to protect space travelers/residents from infectious diseases during long-term flight or space habitation. For these experiments we will: 1: Compare the virulence of space-flight or ground control cultures of S. pneumoniae cultured at ambient temperature or 37oC using intraperitoneal challenge in mice. 2: Investigate global transcriptional patterns of space-flight or ground control cultures of S. pneumoniae cultured at ambient temperature.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The experiments in this study revealed new insights about how opportunistic bacterial pathogens respond to low shear environments in terms of the genes they express and their ability to produce disease. S. pneumoniae is a bacterial pathogen that mediates significant disease in the elderly and the very young. It is also responsible for 40,000 deaths in the US annually. Results from this study advanced our understanding of the pathogenic mechanism of this bacterium and how it adapts to different environments. Additionally, understanding how this microbe responds to low shear environments will allow a rationale risk assessment for this and other opportunistic bacterial pathogens in the space environment. This could be important in the planning and execution of long-term space missions and for the habitation of space. Our results will also lead to development of countermeasures for use on-orbit to detect or treat illnesses on-orbit and on Earth. Finally, information gained about the S. pneumoniae pathogenic mechanism will have direct Earth benefits in terms of understanding the diseases mediated by this microbe and could lead to novel vaccines and therapeutics.
Task Progress:	The protection of long-duration space travelers from infectious organisms remains a major challenge for NASA today. This concern is heightened since the effects of microgravity, other elements of the space environment (ie. radiation, stress, isolation), and the “closed system” nature of spacecraft on the human immune system are well established. Long duration spaceflight has been shown to lead to significant reductions in cellular and humoral immunity which can compromise host defense and lead to problems with infectious disease agents. We believe that opportunistic pathogens represent a significant threat to space travelers. These are microbes that do not cause disease in healthy individuals but can cause significant disease in those that become immune compromised. The Gram-positive bacterium, Streptococcus pneumoniae is an opportunistic pathogen that can cause a spectrum of diseases. S. pneumoniae can lead to middle ear infections, or become life threatening as lobar pneumonia, cross the blood brain barrier to mediate meningitis, become blood borne to cause bacteremia and sepsis. In addition, this microbe is a commensal habitant of the nasopharynx of healthy adults and has been cultured from astronauts prior to flight. Thus, there is the potential for a “perfect storm” - an opportunistic pathogen which is carried aboard by the crew who experience downturns in immune function in space. Add to this accumulating evidence that show that microbial virulence is increased in microgravity environments. In a series of spaceflight experiments (SPEGIS, MVDR, and SPEGIS II), we have investigated changes in the total proteome and global transcriptional activity of S. pneumoniae and the impact of these changes on the virulence of this bacterium. Gene and protein expression experiments were conducted on shuttle missions -STS118 and STS129 and in vivo virulence assessed from cultures flown as part of MVDR on STS123. S. pneumoniae cultures were flown as the SPEGIS experiment during the space shuttle mission - STS118, grown on the International Space Station, and compared to paired ground cultures upon their return to Earth. Cultures were transported to the ISS at 4oC, incubated at 37oC for 15.5hrs and returned to earth at -55oC. Comparison between flight and ground control cultures revealed differences in growth and in the levels of two important S. pneumoniae virulence factors, neuraminidase and ß-galactosidase. Examination of the S. pneumoniae total proteome by two dimensional polyacrylamide electrophoresis revealed approximately twenty protein spot differences with ~75% of these proteins showing enhanced levels after spaceflight with the other 25% showing reduced levels. We also investigated the genes being utilized in the space environment using global transcriptional analysis which showed alterations in more than one hundred expressed genes. These genes represented diverse functional groups and mapped to all areas on the bacterial chromosome. Some of these changes in gene and protein expression associated with space flight are observed in microgravity grown Gram negative bacteria - Salmonella typhimurium. This suggested that bacteria may utilize a common mechanism(s) in responding and adapting to this unique environment. S. pneumoniae cultures were also flown during the space shuttle mission - STS123 as part of the Microbial Virulence Drug Resistance (MVDR) experiment. This used the Bioserve Group Activation Pack (GAP) hardware and fluid processing apparatus (FPA). Cultures were launched and stored at ambient, before activation and returned to earth at ambient temperature. Because of the sub-optimal growth temperature, there was slow growth in all samples. This limited the analysis on the spaceflight samples to biological assays to assess virulence. For flight cultures, there was sufficient growth of the cultures to allow optimal challenge doses to be prepared and utilized. There was an obvious and clear difference in animal virulence between the flight and ground control cultures. Flight cultures were sixteen fold more

	virulent compared to the parallel ground control cultures. This suggests that suboptimal growth in the space environment leads to major changes in S. pneumoniae virulence. However, the overall virulence of these suboptimal cultures were still much higher than that observed for this strain at the higher more optimal culture conditions (37oC). For flight STS129, the Group Activation Pack (GAP) hardware and fluid processing apparatus (FPA) were again used. Cultures were launched and stowed at 6oC before activation. Both room temperature and 37oC cultures were included in the experiment. Room temperature cultures were returned to earth at ambient temperature while 37oC were placed at 6oC for the return to Earth. This experiment allowed comparison of flight and ground samples from all three spaceflight experiments and corresponding ground controls. In addition, statistical analysis allowed comparison to previous gene expression studies conducted in low shear modeled microgravity studies in the lab. Principal component analysis (PCA) for all groups showed that differentially expressed genes observed in low shear modeled microgravity and ground controls clustered together, whereas ground controls and low shear modeled microgravity and flight samples were easily distinguished. Interestingly, there fewer differences in gene expression observed between flight samples grown at 23oC and 37oC. This suggests that space-induced expression changes were temperature independent. In addition, more than twenty virulence-related genes were shown to be altered in response to spaceflight. Interesting, down regulated genes included those involved with protein synthesis and DNA metabolism/repair consistent with a stress response. Additional experiments to examine total protein expression in spaceflight versus ground cultures showed that protein expression differences varied both by temperature and spaceflight versus ground control growth conditions. There were multiple conclusions from these experiments. First, there is a dramatic alteration of global transcription and of the proteome of S. pneumoniae in the space environment. The levels of many genes and proteins were altered. The changes observed in gene expression were independent of culture temperature and reproducible across different missions. Next, there was a surprisingly small overlap with changes seen during low shear modeled microgravity conditions. This suggests there are limitations to this laboratory model in terms of what occurs at the molecular level during spaceflight at least for this bacterium. Next, there was a substantial change in the virulence of S. pneumoniae after growth at sub-optimal temperatures during spaceflight. Flight grown S. pneumoniae are significantly more virulent than parallel cultures ground on Earth. This indicates that the spaceflight environment leads to alterations in gene and protein expression that changes the virulence of this opportunistic pathogen. The basis of regulation of this altered gene/protein expression and virulence activity after spaceflight remains unknown. In Salmonella, the regulator -- Hfq -- appears to be important in the bacterium’s adaptation to the space environment. However, there is no Hfq homologue in S. pneumoniae so the regulation in the space environment is unique. Overall, this study adds to accumulating information that suggests that mitigating infectious disease risks will be important during extended space missions. We believe that these findings have significant implications for long-term space flight or space habitation.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
Abstracts for Journals and Proceedings	Niesel DW, Pandya U, Galindo C, Carmical R, Watson DA, Blell Z, Williams N. "Spaceflight Alters Global Gene/Protein Expression and Enhances the Virulence of Streptococcus pneumoniae." Presented at Joint Meeting of the American Society for Gravitational and Space Biology and International Society for Gravitational Physiology, San Jose, CA, November 2-6, 2011. Program and Abstracts. Joint Meeting of the American Society for Gravitational and Space Biology and International Society for Gravitational Physiology, San Jose, CA, November 2-6, 2011. p. 40. , Nov-2011